

State machines for bounded agents

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Logs traces and privacy

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Evaluation pipelines

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Testing pure components

Without opening the old lesson, explain one key idea from Testing pure components. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for State machines for bounded agents. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for State machines for bounded agents. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Only these transitions are allowed: ready→looking_up→review→done. Done is terminal. Evaluate ready→review→done and ready→looking_up→review→done.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Trace ready -> looking_up -> review -> looking_up -> done. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: An action budget and a transition policy are separate checks; satisfying one does not guarantee the other. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Trace ready -> looking_up -> review -> done -> looking_up. Find first invalid transition.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled state machines for bounded agents evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of state machines for bounded agents. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: State machines for bounded agents

A parcel lookup state machine. An agent may follow idle→querying→checking→complete; complete is terminal.

A follows all four states. B follows idle→querying→complete. No other transitions are declared.

State machines for bounded agents

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read arrows

The graph permits three forward transitions.



2. Follow A

All adjacent pairs match permitted arrows.



3. Inspect B

Querying -> complete skips checking and is invalid.

CHANGE ONE THING

Add checking -> querying as a permitted retry.

Explain the changed result using the matching lesson.

Trace A and B through the state diagram. Name B's forbidden transition and explain whether adding checking→querying permits that shortcut.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A parcel lookup state machine. An agent may follow idle→querying→checking→complete; complete is terminal.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Querying→complete skips checking and is invalid.

1. Follow A
2. Read arrows
3. Inspect B

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Inspect B Querying→complete skips checking and is invalid. A state machine bounds action order. A plausible final state does not establish that the path was valid.

Practice 2: explain the deciding evidence

Changed-case practice: Add checking→querying as a permitted retry. Which conclusion is supported by the supplied facts?

1. That retry becomes valid; querying→complete stays forbidden.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That retry becomes valid; querying→complete stays forbidden. Adding one arrow does not authorise unrelated shortcuts.

Repair a missing building block

Review Logs traces and privacy (ai-6/12). Goal: Explain the mechanism and terminology of logs traces and privacy. Short practice: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?. Return to this week and explain what you changed.

Review Evaluation pipelines (ai-6/10). Goal: Explain the mechanism and terminology of evaluation pipelines. Short practice: Calculate A's and B's correct rates on all four cases. Identify their shared failure and explain why removing that case after scoring changes the honesty of the comparison.. Return to this week and explain what you changed.

Review Testing pure components (ai-6/7). Goal: Explain the mechanism and terminology of testing pure components. Short practice: Predict the formatter outputs for 7 and 12, including spacing. Why does input 0 need rejection under this contract rather than an ordinary formatted response?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
